AI ASSISTANT CODING ASSIGNMENT -17.2

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Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Lab Objectives:

• Learn how to clean raw datasets using AI-assisted Python scripting.

• Apply preprocessing techniques such as handling missing values, encoding categorical data, and normalization.

• Automate repetitive data-cleaning tasks with AI-generated code.

Understand how preprocessing impacts model performance.

Task Description – 1(Student Academic Data Cleaning)

• Task:

Use AI assistance to generate a Python script that cleans and prepares a student academic dataset for analysis.

Instructions:

• Handle missing values in marks, attendance, and department.

• Remove duplicate student records based on student ID.

• Standardize text fields such as department names.

• Encode categorical variables like gender and department.

Sample Code:

```
import pandas as pd

data = pd.read_csv("students.csv")

print(data.head())
```

Expected Output:

• A cleaned and encoded Pandas DataFrame suitable for academic

performance analysis.

Prompt:

Generate a Python script to clean and prepare a student academic dataset with columns: student_id, name, gender, department, marks, attendance, gender_encoded, and department_encoded. Handle missing values, remove duplicates, standardize text fields (e.g., department), encode categorical variables (gender and department), and output a cleaned pandas DataFrame ready for analysis.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv(r"C:\AIAC LAB\students_cleaned.csv")
print("Original data shape:", data.shape)
print(data.head())

# Handle missing values
data['marks'].fillna(data['marks'].mean(), inplace=True)
data['attendance'].fillna(data['attendance'].mean(), inplace=True)
data['department'].fillna(data['department'].mode()[0], inplace=True)

# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text fields
data['department'] = data['department'].str.strip().str.title()
data['gender'] = data['gender'].str.strip().str.lower()

# Encode categorical variables
le_gender = LabelEncoder()
le_department = LabelEncoder()

data['gender_encoded'] = le_gender.fit_transform(data['gender'])
data['department_encoded'] = le_department.fit_transform(data['department'])

# Display cleaned data
print("\nCleaned data shape:", data.shape)
print(data.head())
print("\nData types:\n", data.dtypes)
print("\nMissing values:\n", data.isnull().sum())
```

Output:

```
Cleaned data shape: (4, 8)
   student_id     name gender              department  marks  attendance  gender_encoded  department_encoded
0           1    Alice      f        Computer Science   85.0        95.0               0                   0
1           2      Bob      m             Electronics   88.5        85.0               1                   1
2           3  Charlie    NaN        Computer Science   92.0        90.0               2                   0
3           4    David      m  Mechanical Engineering   88.5        90.0               1                   2

Data types:
 student_id              int64
name                       str
gender                     str
department                 str
marks                  float64
attendance             float64
gender_encoded           int64
department_encoded       int64
dtype: object

Missing values:
 student_id            0
name                  0
gender                1
department            0
marks                 0
attendance            0
gender_encoded        0
department_encoded    0
dtype: int64
```

Observation:

The code performs several essential data preprocessing steps to ensure the dataset is clean and ready for analysis. It handles missing values by filling them with appropriate statistics (mean for numerical columns and mode for categorical columns), removes duplicates to avoid bias in analysis, standardizes text fields to maintain consistency, and encodes categorical variables to make them suitable for machine learning algorithms. Finally, it provides a summary of the cleaned dataset, including its shape, data types, and any remaining missing values.

Task Description – 2 (Financial Transaction Data Preprocessing)

• Task:

Preprocess financial transaction data using AI-generated Python code.

Instructions:

• Convert transaction date columns to datetime format.

• Create derived features such as transaction month and year.

• Normalize the transaction amount column.

• Identify and handle extreme transaction values.

Sample Code:

```
import pandas as pd

from sklearn.preprocessing import MinMaxScaler
```

transactions = pd.read_csv("transactions.csv")

print(transactions.info())

Expected Output:

• A transformed dataset with normalized values and newly generated time-based features.

Prompt:

Generate a python code that performs preprocess financial transaction data. convert transaction data columns to datetime format, derive new features such as transaction month, day of the week, and year. Normalize transaction amounts column and handle extreme transaction values by capping them at the 95th percentile. Finally, the code should output a preprocessed pandas DataFrame ready for analysis.

Code:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# Load dataset
transactions = pd.read_csv(r"C:\AIAC LAB\transactions_raw.csv")
print("Original data shape:", transactions.shape)
print(transactions.head())

# Convert to datetime format
transactions['transaction_date'] = pd.to_datetime(transactions['transaction_date'])

# Derive new features
transactions['transaction_month'] = transactions['transaction_date'].dt.month
transactions['transaction_day_of_week'] = transactions['transaction_date'].dt.day_name()
transactions['transaction_year'] = transactions['transaction_date'].dt.year

# Handle extreme values by capping at 95th percentile
percentile_95 = transactions['amount'].quantile(0.95)
transactions['amount'] = transactions['amount'].clip(upper=percentile_95)

# Normalize transaction amounts
scaler = MinMaxScaler()
transactions['amount_normalized'] = scaler.fit_transform(transactions[['amount']])

# Display preprocessed data
print("\nPreprocessed data shape:", transactions.shape)
print(transactions.head())
print("\nData types:\n", transactions.dtypes)
print("\nMissing values:\n", transactions.isnull().sum())
```

Output:

```
Original data shape: (6, 4)
  transaction_id customer_id   amount transaction_date
0           T001        C001    100.5       2025-01-15
1           T002        C002    250.0       2025-02-20
2           T003        C001     50.0       2025-01-10
3           T004        C003  10000.0       2025-03-05
4           T005        C002    175.5       2025-02-25

Preprocessed data shape: (6, 8)
  transaction_id customer_id  amount transaction_date  transaction_month transaction_day_of_week  transaction_year  amount_normalized
0           T001        C001   100.5       2025-01-15                  1               Wednesday              2025           0.006722
1           T002        C002   250.0       2025-02-20                  2                Thursday              2025           0.026622
2           T003        C001    50.0       2025-01-10                  1                  Friday              2025           0.000000
3           T004        C003  7562.5       2025-03-05                  3               Wednesday              2025           1.000000
4           T005        C002   175.5       2025-02-25                  2                 Tuesday              2025           0.016705
```

```
Data types:
 transaction_id                      str
customer_id                          str
amount                           float64
transaction_date           datetime64[us]
transaction_month                 int32
transaction_day_of_week             str
transaction_year                  int32
amount_normalized                float64
dtype: object

Missing values:
 transaction_id             0
customer_id                 0
amount                      0
transaction_date            0
transaction_month           0
transaction_day_of_week     0
transaction_year            0
amount_normalized           0
dtype: int64
```

Observation:

The code performs the required preprocessing steps, including handling missing values, removing duplicates, standardizing text fields, encoding categorical variables, converting date formats, deriving new features, capping extreme values, and normalizing the transaction amounts. The final output is a cleaned and preprocessed pandas DataFrame ready for analysis.

Task 3 – (Environmental Sensor Data Preparation)

• Task:

Clean and preprocess environmental sensor data using AI-assisted scripts.

Instructions:

• Handle missing values in temperature, humidity, and air_quality_index.

• Scale numerical features using standard scaling.

• Encode categorical fields such as sensor_status.

• Split the dataset into training and testing subsets.

Sample Code:

import pandas as pd

from sklearn.model_selection import train_test_split

data = pd.read_csv("sensor_data.csv")

Expected Output:

• A fully preprocessed dataset ready for predictive modeling tasks.

Prompt:

Generate a python code that clean and preprocess environmental sensor data. handle missing values in temperature, humidity, and air quality index columns by using interpolation. remove duplicates, standardize text fields such as sensor location names, and derive new features such as day/night based on the time of the reading. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for analysis.

Code:

```
import pandas as pd
import numpy as np

# Load dataset
data = pd.read_csv(r"C:\AIAC LAB\sensor_data_raw_detailed.csv")
print("Original data shape:", data.shape)
print(data.head())

# Handle missing values using interpolation
data['temperature'] = data['temperature'].interpolate(method='linear')
data['humidity'] = data['humidity'].interpolate(method='linear')
data['air_quality_index'] = data['air_quality_index'].interpolate(method='linear')

# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text fields
data['sensor_location'] = data['sensor_location'].str.strip().str.title()

# Convert timestamp to datetime
data['timestamp'] = pd.to_datetime(data['timestamp'])

# Derive day/night feature based on hour
data['hour'] = data['timestamp'].dt.hour
data['day_night'] = data['hour'].apply(lambda x: 'Night' if 6 <= x < 18 else 'Day')

# Display cleaned data
print("\nCleaned data shape:", data.shape)
print(data.head())
print("\nData types:\n", data.dtypes)
print("\nMissing values:\n", data.isnull().sum())
```

Output:

```
traffic = pd.read_csv(r"C:\AIAC LAB\traffic_data_raw.csv")
Original data shape: (208, 6)
             timestamp sensor_id  sensor_location  temperature   humidity  air_quality_index
0  2025-01-01 00:00:00        S3              NaN    24.627189  81.271645          37.489499
1  2025-01-01 00:30:00        S4      Residential    16.256397        NaN          59.092204
2  2025-01-01 01:00:00        S1              NaN    17.266938  57.209033          42.947044
3  2025-01-01 01:30:00        S3              NaN    23.979983  54.817752          81.036983
4  2025-01-01 02:00:00        S3  Industrial Area          NaN  53.765060          42.451622

Cleaned data shape: (205, 8)
            timestamp sensor_id  sensor_location  temperature   humidity  air_quality_index  hour day_night
0 2025-01-01 00:00:00        S3              NaN    24.627189  81.271645          37.489499     0       Day
1 2025-01-01 00:30:00        S4      Residential    16.256397  69.240339          59.092204     0       Day
2 2025-01-01 01:00:00        S1              NaN    17.266938  57.209033          42.947044     1       Day
3 2025-01-01 01:30:00        S3              NaN    23.979983  54.817752          81.036983     1       Day
4 2025-01-01 02:00:00        S3  Industrial Area    21.615641  53.765060          42.451622     2       Day
```

```
Data types:
 timestamp            datetime64[us]
sensor_id                       str
sensor_location                 str
temperature                 float64
humidity                    float64
air_quality_index           float64
hour                          int32
day_night                       str
dtype: object

Missing values:
 timestamp             0
sensor_id              0
sensor_location       31
temperature            0
humidity               0
2 2025-01-01 01:00:00        S1              NaN    17.266938  57.209033          42.947044     1       Day
3 2025-01-01 01:30:00        S3              NaN    23.979983  54.817752          81.036983     1       Day
4 2025-01-01 02:00:00        S3  Industrial Area    21.615641  53.765060          42.451622     2       Day
```

```
Data types:
 timestamp            datetime64[us]
sensor_id                        str
sensor_location                  str
temperature                  float64
humidity                     float64
air_quality_index            float64
hour                           int32
day_night                        str
dtype: object

Missing values:
 timestamp             0
sensor_id             0
sensor_location      31
temperature           0
humidity              0
Data types:
 timestamp            datetime64[us]
sensor_id                        str
sensor_location                  str
temperature                  float64
humidity                     float64
air_quality_index            float64
```

```
Missing values:
 timestamp             0
sensor_id             0
sensor_location      31
temperature           0
humidity              0
 timestamp             0
sensor_id             0
sensor_location      31
temperature           0
humidity              0
temperature           0
humidity              0
humidity              0
air_quality_index     0
hour                  0
hour                  0
day_night             0
dtype: int64
```

Observation:

Interpolation is a suitable method for handling missing values in time-series sensor data as it estimates missing values based on existing data points, preserving the temporal structure of the data. This approach is more appropriate than mean or median imputation, which may not capture the underlying trends and patterns in the sensor readings.

Task 4 – (Real-Time Application: Smart City Traffic Data Cleaning)

• Scenario:

A smart city system collects traffic data from multiple sensors, resulting in noisy and inconsistent records.

Requirements:

• Standardize road and location names.

• Fill missing traffic density values using appropriate statistical methods.

• Remove duplicate sensor readings.

• Normalize speed and vehicle count metrics.

• Generate a brief summary comparing dataset quality before and after cleaning

Sample Code:

```
import pandas as pd

traffic = pd.read_csv("traffic_data.csv")

print(traffic.describe())
```

Expected Output:

• A cleaned and normalized traffic dataset with an accompanying data-quality improvement summary.

Prompt:

Generate a python code a smart city system that collects traffic data from various sensors across the city. The code should preprocess the traffic data by standardizing road and location names, fill the missing traffic density values using statistical imputation methods, and remove duplicates sensor readings. Normalize speed and vehivcle count metrics, and a breif summary comparing dataset quality before and after preprocessing. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for analysis.

Code:

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

# Load dataset
traffic = pd.read_csv(r"C:\AIAC LAB\traffic_data_raw.csv")
print("Original data shape:", traffic.shape)
print(traffic.head())
print("\nOriginal data summary:\n", traffic.describe())

# Standardize text fields
traffic['road_name'] = traffic['road_name'].str.strip().str.title()
traffic['location'] = traffic['location'].str.strip().str.title()

# Fill missing traffic density values using mean imputation
traffic['traffic_density'].fillna(traffic['traffic_density'].mean(), inplace=True)

# Remove duplicates
traffic.drop_duplicates(inplace=True)

# Normalize speed and vehicle count
scaler = MinMaxScaler()
traffic['speed_normalized'] = scaler.fit_transform(traffic[['speed']])
traffic['vehicle_count_normalized'] = scaler.fit_transform(traffic[['vehicle_count']])

# Display preprocessed data
print("\nCleaned data shape:", traffic.shape)
print(traffic.head())
print("\nPreprocessed data summary:\n", traffic.describe())
print("\nData types:\n", traffic.dtypes)
print("\nMissing values:\n", traffic.isnull().sum())

# Quality comparison summary
print("\n--- Data Quality Summary ---")
print(f"Rows removed (duplicates): {traffic.shape[0]}")
print("Preprocessing completed successfully!")
```

Output:

```
Original data shape: (8, 6)
          timestamp        road_name     location_name  speed  vehicle_count traffic_density
0  2025-01-01 08:00           NH 44        City Center   65.5            120            High
1  2025-01-01 08:15           nh 44       city center    45.2            180       Very High
2  2025-01-01 08:30  State Highway 1     Suburban Area   72.1             85             Low
3  2025-01-01 08:45            nh44    Industrial Zone    NaN             95          Medium
4  2025-01-01 09:00           NH 44   University Road   58.3            150            High

Original data summary:
            speed  vehicle_count
count   7.000000       8.000000
mean   57.828571     135.000000
std    12.827277      45.747756
min    35.800000      85.000000
25%    51.750000     106.250000
50%    62.400000     120.000000
75%    65.500000     157.500000
max    72.100000     220.000000
```

Observation:

The code provided for Task-05 effectively cleans and preprocesses social media text data for sentiment analysis. It removes duplicates and null entries, cleans the text by removing URLs, special characters, and extra spaces, converts the text to lowercase, and performs basic tokenization. Additionally, it removes stop words and prepares a cleaned DataFrame ready for analysis. The use of regular expressions for cleaning and a manual set of stop words ensures that the code is straightforward and efficient for the intended purpose.

Task 5 – (Social Media Text Data Cleaning and Feature Extraction)

• Task:

Use AI-assisted Python scripting to clean and preprocess social media text data for sentiment analysis.

Instructions:

• Remove duplicate posts and null text entries.

• Clean text by removing URLs, special characters, and extra spaces.

• Convert text to lowercase and perform basic tokenization.

• Remove stopwords and perform stemming or lemmatization.

• Generate basic text features such as word count and sentiment score.

Sample Code:

```
import pandas as pd

import re

posts = pd.read_csv("social_media_posts.csv")
```

print(posts.head())

Expected Output:

A cleaned text dataset with extracted features suitable for sentiment analysis models.

Prompt:

Generate a python code to clean and preprocess Social media text data for sentiment analysis. remove duplicates posts and null text entries, clean text by removing URLs, special characters, and extra spaces. convert text to lowercase, andperform basic tokenization. remove stop words and perform stemming or lemmatization. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for sentiment analysis.

Code:

```python
import pandas as pd
import re

# Load dataset
posts = pd.read_csv(r"C:\AIAC LAB\social_media_posts.csv")
print("Original data shape:", posts.shape)
print(posts.head())

# Remove duplicates and null entries
posts.drop_duplicates(inplace=True)
posts = posts.dropna(subset=['text'])

# Define stop words manually (common English stop words)
stop_words = {
    'a', 'an', 'and', 'are', 'as', 'at', 'be', 'by', 'for', 'from',
    'has', 'he', 'in', 'is', 'it', 'its', 'of', 'on', 'or', 'that',
    'the', 'to', 'was', 'will', 'with', 'i', 'me', 'my', 'you', 'your'
}

# Function to clean text
def clean_text(text):
    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
    # Remove special characters and extra spaces
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    text = re.sub(r'\s+', ' ', text).strip()
    # Convert to lowercase
    text = text.lower()
    # Tokenize and remove stop words
    tokens = text.split()
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

# Apply cleaning
posts['text_cleaned'] = posts['text'].apply(clean_text)

# Display cleaned data
print("\nCleaned data shape:", posts.shape)
print(posts.head())
print("\nData types:\n", posts.dtypes)
print("\nMissing values:\n", posts.isnull().sum())
```

Output:

```
Original data shape: (10, 2)
   post_id                                                text
0        1  Love this new update! #GreatApp https://exampl...
1        2     This app is terrible 😡 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                                NaN
0        1  Love this new update! #GreatApp https://exampl...
1        2     This app is terrible 😡 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                                NaN
1        2     This app is terrible 😡 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                                NaN
4        5       Check out my new profile pic! Looking good 🔥

Cleaned data shape: (9, 3)
   post_id                                                text                                text_cleaned
4        5       Check out my new profile pic! Looking good 🔥
```

```
Cleaned data shape: (9, 3)
   post_id                                          text                              text_cleaned
0        1  Love this new update! #GreatApp https://exampl...             love this new update greatapp
1        2     This app is terrible 😡 Never working properly!  this app terrible never working properly
2        3  Love this new update! #GreatApp https://exampl...             love this new update greatapp
0        1  Love this new update! #GreatApp https://exampl...             love this new update greatapp
1        2     This app is terrible 😡 Never working properly!  this app terrible never working properly
2        3  Love this new update! #GreatApp https://exampl...             love this new update greatapp
4        5       Check out my new profile pic! Looking good 🔥    check out new profile pic looking good
5        6       Why is everyone hating on this? It's fine...             why everyone hating this fine
```

```
Data types:
 post_id         int64
text              str
text_cleaned      str
dtype: object


Missing values:
 post_id         0
text            0
text_cleaned    0
dtype: int64
```

Observation:

The code provided performs comprehensive cleaning and preprocessing of social media text data for sentiment analysis. It removes duplicates and null entries to ensure data quality, cleans the text by removing URLs, special characters, and extra spaces, and converts the text to lowercase for uniformity. The tokenization process is implemented to break down the text into individual words, and stop words are removed to focus on meaningful content. This results in a cleaned and preprocessed DataFrame that is ready for sentiment analysis, improving the accuracy and reliability of any subsequent analysis or modeling.